

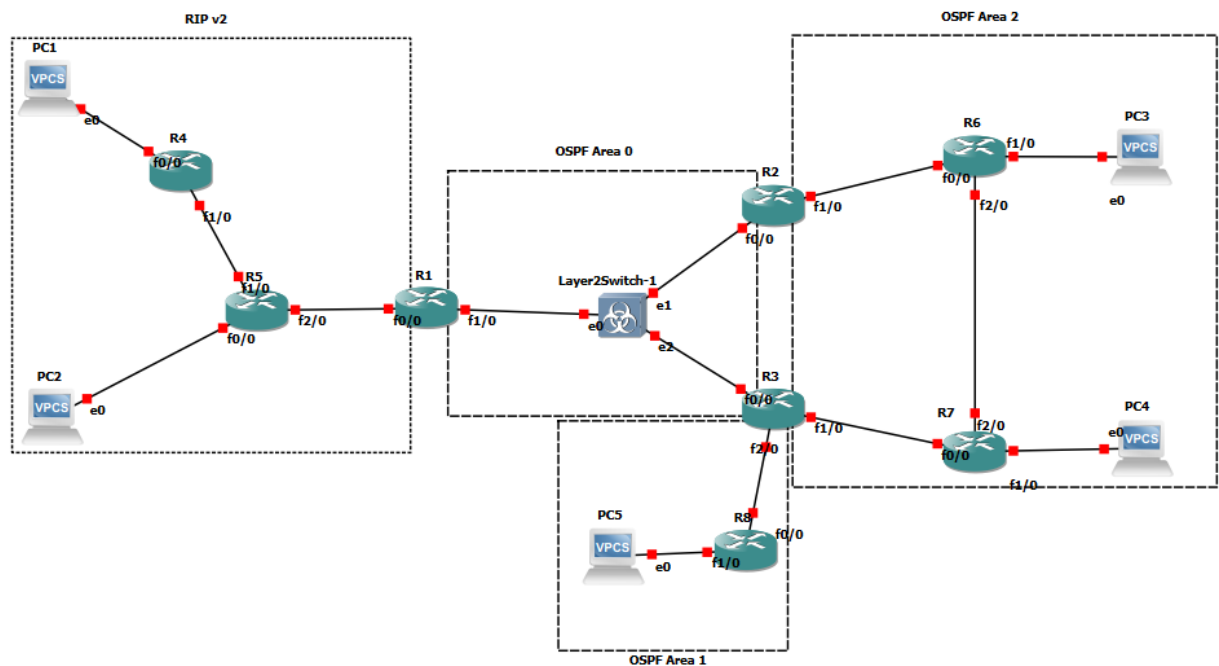


Рисунок 1 – Собранная схема

- Диапазон 192.168.x.x — для сегмента RIP v2.
- Диапазон 10.0.0.0/24 — для зоны OSPF Area 0 (мультидоступный сегмент через коммутатор Layer2Switch-1).
- Диапазон 10.1.x.x — для OSPF Area 1 (тупиковая зона).
- Диапазон 10.2.x.x — для OSPF Area 2.

Таблица подсетей и портов:

- Сеть PC1 – R4: 192.168.14.0/24
- Сеть PC2 – R5: 192.168.15.0/24
- Сеть R4 – R5: 192.168.45.0/30 (маска /30 или 255.255.255.252 идеальна для стыков «точка-точка», так как содержит всего 2 полезных IP-адреса)
- Сеть R5 – R1: 192.168.51.0/30

- Сеть Area 0 (R1 – R2 – R3): 10.0.0.0/24 (все три роутера подключены к одному коммутатору)
- Сеть R3 – R8 (Area 1): 10.1.38.0/30
- Сеть R8 – PC5 (Area 1): 10.1.85.0/24
- Сеть R2 – R6 (Area 2): 10.2.26.0/30
- Сеть R3 – R7 (Area 2): 10.2.37.0/30
- Сеть R6 – R7 (Area 2): 10.2.67.0/30
- Сеть R6 – PC3 (Area 2): 10.2.63.0/24
- Сеть R7 – PC4 (Area 2): 10.2.74.0/24

PC1:

```
ip 192.168.14.10/24 192.168.14.1
```

```
save
```

PC2:

```
ip 192.168.15.10/24 192.168.15.1
```

```
save
```

PC3:

```
ip 10.2.63.10/24 10.2.63.1
```

```
save
```

PC4:

```
ip 10.2.74.10/24 10.2.74.1
```

```
save
```

PC5:

```
ip 10.1.85.10/24 10.1.85.1
```

```
save
```

R1:

```
enable
```

```
configure terminal
```

```
interface FastEthernet0/0
ip address 192.168.51.2 255.255.255.252
no shutdown
exit
```

```
interface FastEthernet1/0
ip address 10.0.0.1 255.255.255.0
no shutdown
exit
```

```
R2:
enable
configure terminal
```

```
interface FastEthernet0/0
ip address 10.0.0.2 255.255.255.0
no shutdown
exit
```

```
interface FastEthernet1/0
ip address 10.2.26.1 255.255.255.252
no shutdown
exit
```

```
R3:
enable
configure terminal
```

```
interface FastEthernet0/0
ip address 10.0.0.3 255.255.255.0
no shutdown
exit
```

```
interface FastEthernet1/0
ip address 10.2.37.1 255.255.255.252
no shutdown
exit
```

```
interface FastEthernet2/0
ip address 10.1.38.1 255.255.255.252
no shutdown
exit
```

```
R4:
enable
configure terminal
```

```
interface FastEthernet0/0
ip address 192.168.14.1 255.255.255.0
no shutdown
exit
```

```
interface FastEthernet1/0
ip address 192.168.45.1 255.255.255.252
no shutdown
exit
```

R5:

enable

configure terminal

interface FastEthernet0/0

ip address 192.168.15.1 255.255.255.0

no shutdown

exit

interface FastEthernet1/0

ip address 192.168.45.2 255.255.255.252

no shutdown

exit

interface FastEthernet2/0

ip address 192.168.51.1 255.255.255.252

no shutdown

exit

R6:

enable

configure terminal

interface FastEthernet0/0

ip address 10.2.26.2 255.255.255.252

no shutdown

exit

interface FastEthernet1/0

```
ip address 10.2.63.1 255.255.255.0
no shutdown
exit
```

```
interface FastEthernet2/0
ip address 10.2.67.1 255.255.255.252
no shutdown
exit
```

```
R7:
enable
configure terminal
```

```
interface FastEthernet0/0
ip address 10.2.37.2 255.255.255.252
no shutdown
exit
```

```
interface FastEthernet1/0
ip address 10.2.74.1 255.255.255.0
no shutdown
exit
```

```
interface FastEthernet2/0
ip address 10.2.67.2 255.255.255.252
no shutdown
exit
```

```
R8:
```

```
enable
configure terminal
interface FastEthernet0/0
ip address 10.1.38.2 255.255.255.252
no shutdown
exit
```

```
interface FastEthernet1/0
ip address 10.1.85.1 255.255.255.0
no shutdown
exit
```

```
PC1> ping 192.168.14.1

84 bytes from 192.168.14.1 icmp_seq=1 ttl=255 time=20.154 ms
84 bytes from 192.168.14.1 icmp_seq=2 ttl=255 time=10.729 ms
84 bytes from 192.168.14.1 icmp_seq=3 ttl=255 time=9.911 ms
84 bytes from 192.168.14.1 icmp_seq=4 ttl=255 time=1.939 ms
84 bytes from 192.168.14.1 icmp_seq=5 ttl=255 time=4.935 ms
```

Рисунок 2 – С консоли PC1 пинг до его шлюза

```
R4#ping 192.168.45.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.45.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/57/64 ms
R4#
```

Рисунок 3 – С консоли роутера R4 пинг до R5

```
R1#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/60/64 ms
R1#ping 10.0.0.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.3, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/58/60 ms
R1#
```

Рисунок 4 – С консоли роутера R1 доступность соседей в Area 0

Настройка R4:

```
configure terminal
router rip
version 2
no auto-summary
network 192.168.14.0
network 192.168.45.0
exit
```

Настройка R5:

```
configure terminal
router rip
version 2
no auto-summary
network 192.168.15.0
network 192.168.45.0
network 192.168.51.0
exit
```

Настройка R1:

```
configure terminal
router rip
version 2
no auto-summary
network 192.168.51.0
exit
```


show ip route rip

```
R4#  
R4#show ip route rip  
R    192.168.15.0/24 [120/1] via 192.168.45.2, 00:00:26, FastEthernet1/0  
    192.168.51.0/30 is subnetted, 1 subnets  
R        192.168.51.0 [120/1] via 192.168.45.2, 00:00:26, FastEthernet1/0  
R4#
```

Рисунок 5 – Таблица маршрутизации

```
PC1>  
PC1> ping 192.168.15.10  
  
84 bytes from 192.168.15.10 icmp_seq=1 ttl=62 time=26.124 ms  
84 bytes from 192.168.15.10 icmp_seq=2 ttl=62 time=24.243 ms  
84 bytes from 192.168.15.10 icmp_seq=3 ttl=62 time=22.974 ms  
84 bytes from 192.168.15.10 icmp_seq=4 ttl=62 time=26.465 ms  
84 bytes from 192.168.15.10 icmp_seq=5 ttl=62 time=42.916 ms  
  
PC1>
```

Рисунок 6 – PC1 пинг до PC2

```
PC1>  
PC1> ping 192.168.51.2  
  
84 bytes from 192.168.51.2 icmp_seq=1 ttl=253 time=26.323 ms  
84 bytes from 192.168.51.2 icmp_seq=2 ttl=253 time=26.971 ms  
84 bytes from 192.168.51.2 icmp_seq=3 ttl=253 time=21.037 ms  
84 bytes from 192.168.51.2 icmp_seq=4 ttl=253 time=23.613 ms  
84 bytes from 192.168.51.2 icmp_seq=5 ttl=253 time=22.502 ms  
  
PC1>
```

Рисунок 7 – PC1 пинг RIP-интерфейса роутера R1

Настройка R1:

configure terminal

router ospf 1

network 10.0.0.0 0.0.0.255 area 0

exit

Настройка R2:

configure terminal

router ospf 1

```
network 10.0.0.0 0.0.0.255 area 0
network 10.2.26.0 0.0.0.3 area 2
exit
```

Настройка R3:

```
configure terminal
router ospf 1
network 10.0.0.0 0.0.0.255 area 0
network 10.2.37.0 0.0.0.3 area 2
network 10.1.38.0 0.0.0.3 area 1
area 1 stub no-summary
exit
```

Настройка R8:

```
configure terminal
router ospf 1
network 10.1.38.0 0.0.0.3 area 1
network 10.1.85.0 0.0.0.255 area 1
area 1 stub
exit
```

Настройка R6:

```
configure terminal
router ospf 1
network 10.2.26.0 0.0.0.3 area 2
network 10.2.63.0 0.0.0.255 area 2
network 10.2.67.0 0.0.0.3 area 2
exit
```

Настройка R7:

configure terminal

router ospf 1

network 10.2.37.0 0.0.0.3 area 2

network 10.2.74.0 0.0.0.255 area 2

network 10.2.67.0 0.0.0.3 area 2

exit

show ip ospf neighbor

```
R3#
R3#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
10.2.26.1        1     FULL/BDR        00:00:32    10.0.0.2       FastEthernet0/
0
192.168.51.2     1     FULL/DR         00:00:37    10.0.0.1       FastEthernet0/
0
10.1.85.1        1     FULL/BDR        00:00:30    10.1.38.2      FastEthernet2/
0
10.2.74.1        1     FULL/BDR        00:00:33    10.2.37.2      FastEthernet1/
0
R3#
R3#
R3#
```

Рисунок 8 – Проверка соседства

show ip route

```
R8#
R8#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.1.38.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.38.0/30 is directly connected, FastEthernet0/0
C       10.1.85.0/24 is directly connected, FastEthernet1/0
O*IA 0.0.0.0/0 [110/2] via 10.1.38.1, 00:03:30, FastEthernet0/0
R8#
R8#
```

Рисунок 9 – Проверка Totally Stubby Area на R8

ping 10.1.85.10

```
PC3> ping 10.1.85.10

84 bytes from 10.1.85.10 icmp_seq=1 ttl=60 time=51.141 ms
84 bytes from 10.1.85.10 icmp_seq=2 ttl=60 time=41.816 ms
84 bytes from 10.1.85.10 icmp_seq=3 ttl=60 time=42.461 ms
84 bytes from 10.1.85.10 icmp_seq=4 ttl=60 time=62.327 ms
84 bytes from 10.1.85.10 icmp_seq=5 ttl=60 time=50.181 ms

PC3>
```

Рисунок 10 – Проверка пинга внутри зоны OSPF

Настройки на роутере R1:

```
configure terminal
```

```
router rip
```

```
redistribute ospf 1 metric 5
```

```
exit
```

```
router ospf 1
```

```
redistribute rip subnets
```

```
exit
```

Проверяем сторону RIP (на роутере R4):

```
show ip route
```

```

R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.14.0/24 is directly connected, FastEthernet0/0
R    192.168.15.0/24 [120/1] via 192.168.45.2, 00:00:07, FastEthernet1/0
    192.168.45.0/30 is subnetted, 1 subnets
C    192.168.45.0 is directly connected, FastEthernet1/0
    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
R    10.0.0.0/24 [120/6] via 192.168.45.2, 00:00:07, FastEthernet1/0
R    10.2.26.0/30 [120/6] via 192.168.45.2, 00:00:07, FastEthernet1/0
R    10.2.37.0/30 [120/6] via 192.168.45.2, 00:00:07, FastEthernet1/0
R    10.1.38.0/30 [120/6] via 192.168.45.2, 00:00:08, FastEthernet1/0
R    10.2.63.0/24 [120/6] via 192.168.45.2, 00:00:08, FastEthernet1/0
R    10.2.74.0/24 [120/6] via 192.168.45.2, 00:00:08, FastEthernet1/0
R    10.2.67.0/30 [120/6] via 192.168.45.2, 00:00:08, FastEthernet1/0
R    10.1.85.0/24 [120/6] via 192.168.45.2, 00:00:08, FastEthernet1/0
    192.168.51.0/30 is subnetted, 1 subnets
R    192.168.51.0 [120/1] via 192.168.45.2, 00:00:11, FastEthernet1/0
R4#
R4#

```

Рисунок 11 – Сторона рип

Проверяем сторону OSPF (на роутере R6):

show ip route

```

R6#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

O E2 192.168.14.0/24 [110/20] via 10.2.26.1, 00:04:29, FastEthernet0/0
O E2 192.168.15.0/24 [110/20] via 10.2.26.1, 00:04:29, FastEthernet0/0
    192.168.45.0/30 is subnetted, 1 subnets
O E2    192.168.45.0 [110/20] via 10.2.26.1, 00:04:29, FastEthernet0/0
    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
O IA    10.0.0.0/24 [110/2] via 10.2.26.1, 00:13:35, FastEthernet0/0
C        10.2.26.0/30 is directly connected, FastEthernet0/0
O        10.2.37.0/30 [110/2] via 10.2.67.2, 00:13:35, FastEthernet2/0
O IA    10.1.38.0/30 [110/3] via 10.2.67.2, 00:13:36, FastEthernet2/0
        [110/3] via 10.2.26.1, 00:13:36, FastEthernet0/0
C        10.2.63.0/24 is directly connected, FastEthernet1/0
O        10.2.74.0/24 [110/2] via 10.2.67.2, 00:13:36, FastEthernet2/0
C        10.2.67.0/30 is directly connected, FastEthernet2/0
O IA    10.1.85.0/24 [110/4] via 10.2.67.2, 00:13:37, FastEthernet2/0
        [110/4] via 10.2.26.1, 00:13:37, FastEthernet0/0
    192.168.51.0/30 is subnetted, 1 subnets
O E2    192.168.51.0 [110/20] via 10.2.26.1, 00:04:33, FastEthernet0/0
R6#

```

Рисунок 11 – Сторона оспф

Проверка тупиковой зоны (на роутере R8):

show ip route

```

R8#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.1.38.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.1.38.0/30 is directly connected, FastEthernet0/0
C        10.1.85.0/24 is directly connected, FastEthernet1/0
O*IA 0.0.0.0/0 [110/2] via 10.1.38.1, 00:16:39, FastEthernet0/0
R8#
R8#

```

Рисунок 12 – Тупиковая зона проверка

ping 192.168.14.10

ping 192.168.15.10

ping 10.2.63.10

ping 10.2.74.10

ping 10.1.85.10

```
PC1> ping 192.168.14.10

192.168.14.10 icmp_seq=1 ttl=64 time=0.001 ms
192.168.14.10 icmp_seq=2 ttl=64 time=0.001 ms
192.168.14.10 icmp_seq=3 ttl=64 time=0.001 ms
192.168.14.10 icmp_seq=4 ttl=64 time=0.001 ms
192.168.14.10 icmp_seq=5 ttl=64 time=0.001 ms

PC1> ping 192.168.15.10

84 bytes from 192.168.15.10 icmp_seq=1 ttl=62 time=49.787 ms
84 bytes from 192.168.15.10 icmp_seq=2 ttl=62 time=35.304 ms
84 bytes from 192.168.15.10 icmp_seq=3 ttl=62 time=23.105 ms
84 bytes from 192.168.15.10 icmp_seq=4 ttl=62 time=31.065 ms
84 bytes from 192.168.15.10 icmp_seq=5 ttl=62 time=28.245 ms

PC1> ping 10.2.63.10

84 bytes from 10.2.63.10 icmp_seq=1 ttl=59 time=69.441 ms
84 bytes from 10.2.63.10 icmp_seq=2 ttl=59 time=71.283 ms
84 bytes from 10.2.63.10 icmp_seq=3 ttl=59 time=71.437 ms
84 bytes from 10.2.63.10 icmp_seq=4 ttl=59 time=65.240 ms
84 bytes from 10.2.63.10 icmp_seq=5 ttl=59 time=61.633 ms

PC1> ping 10.2.74.10

10.2.74.10 icmp_seq=1 timeout
84 bytes from 10.2.74.10 icmp_seq=2 ttl=59 time=65.991 ms
84 bytes from 10.2.74.10 icmp_seq=3 ttl=59 time=55.793 ms
84 bytes from 10.2.74.10 icmp_seq=4 ttl=59 time=62.103 ms
84 bytes from 10.2.74.10 icmp_seq=5 ttl=59 time=71.391 ms

PC1> ping 10.1.85.10

84 bytes from 10.1.85.10 icmp_seq=1 ttl=59 time=76.004 ms
84 bytes from 10.1.85.10 icmp_seq=2 ttl=59 time=63.248 ms
84 bytes from 10.1.85.10 icmp_seq=3 ttl=59 time=52.402 ms
84 bytes from 10.1.85.10 icmp_seq=4 ttl=59 time=54.787 ms
84 bytes from 10.1.85.10 icmp_seq=5 ttl=59 time=78.806 ms

PC1>
```

Рисунок 13 – Пинги с ПК1

```
PC2> ping 192.168.14.10

84 bytes from 192.168.14.10 icmp_seq=1 ttl=62 time=31.282 ms
84 bytes from 192.168.14.10 icmp_seq=2 ttl=62 time=32.220 ms
84 bytes from 192.168.14.10 icmp_seq=3 ttl=62 time=20.645 ms
84 bytes from 192.168.14.10 icmp_seq=4 ttl=62 time=22.209 ms
84 bytes from 192.168.14.10 icmp_seq=5 ttl=62 time=32.268 ms

PC2> ping 192.168.15.10

192.168.15.10 icmp_seq=1 ttl=64 time=0.001 ms
192.168.15.10 icmp_seq=2 ttl=64 time=0.001 ms
192.168.15.10 icmp_seq=3 ttl=64 time=0.001 ms
192.168.15.10 icmp_seq=4 ttl=64 time=0.001 ms
192.168.15.10 icmp_seq=5 ttl=64 time=0.001 ms

PC2> ping 10.2.63.10

84 bytes from 10.2.63.10 icmp_seq=1 ttl=60 time=60.094 ms
84 bytes from 10.2.63.10 icmp_seq=2 ttl=60 time=54.680 ms
84 bytes from 10.2.63.10 icmp_seq=3 ttl=60 time=51.277 ms
84 bytes from 10.2.63.10 icmp_seq=4 ttl=60 time=53.063 ms
84 bytes from 10.2.63.10 icmp_seq=5 ttl=60 time=46.063 ms

PC2> ping 10.2.74.10

84 bytes from 10.2.74.10 icmp_seq=1 ttl=60 time=53.439 ms
84 bytes from 10.2.74.10 icmp_seq=2 ttl=60 time=55.867 ms
84 bytes from 10.2.74.10 icmp_seq=3 ttl=60 time=43.206 ms
84 bytes from 10.2.74.10 icmp_seq=4 ttl=60 time=50.324 ms
84 bytes from 10.2.74.10 icmp_seq=5 ttl=60 time=51.418 ms

PC2> ping 10.1.85.10

84 bytes from 10.1.85.10 icmp_seq=1 ttl=60 time=48.902 ms
84 bytes from 10.1.85.10 icmp_seq=2 ttl=60 time=43.012 ms
84 bytes from 10.1.85.10 icmp_seq=3 ttl=60 time=65.047 ms
84 bytes from 10.1.85.10 icmp_seq=4 ttl=60 time=62.224 ms
84 bytes from 10.1.85.10 icmp_seq=5 ttl=60 time=42.629 ms

PC2> 
```

Рисунок 14 – Пинги с ПК2


```
PC3> ping 192.168.14.10

84 bytes from 192.168.14.10 icmp_seq=1 ttl=59 time=81.211 ms
84 bytes from 192.168.14.10 icmp_seq=2 ttl=59 time=73.316 ms
84 bytes from 192.168.14.10 icmp_seq=3 ttl=59 time=74.092 ms
84 bytes from 192.168.14.10 icmp_seq=4 ttl=59 time=56.941 ms
84 bytes from 192.168.14.10 icmp_seq=5 ttl=59 time=65.573 ms

PC3> ping 192.168.15.10

84 bytes from 192.168.15.10 icmp_seq=1 ttl=60 time=59.053 ms
84 bytes from 192.168.15.10 icmp_seq=2 ttl=60 time=65.096 ms
84 bytes from 192.168.15.10 icmp_seq=3 ttl=60 time=51.394 ms
84 bytes from 192.168.15.10 icmp_seq=4 ttl=60 time=54.739 ms
84 bytes from 192.168.15.10 icmp_seq=5 ttl=60 time=53.722 ms

PC3> ping 10.2.63.10

10.2.63.10 icmp_seq=1 ttl=64 time=0.001 ms
10.2.63.10 icmp_seq=2 ttl=64 time=0.001 ms
10.2.63.10 icmp_seq=3 ttl=64 time=0.001 ms
10.2.63.10 icmp_seq=4 ttl=64 time=0.001 ms
10.2.63.10 icmp_seq=5 ttl=64 time=0.001 ms

PC3> ping 10.2.74.10

84 bytes from 10.2.74.10 icmp_seq=1 ttl=62 time=31.796 ms
84 bytes from 10.2.74.10 icmp_seq=2 ttl=62 time=21.959 ms
84 bytes from 10.2.74.10 icmp_seq=3 ttl=62 time=22.168 ms
84 bytes from 10.2.74.10 icmp_seq=4 ttl=62 time=22.525 ms
84 bytes from 10.2.74.10 icmp_seq=5 ttl=62 time=24.156 ms

PC3> ping 10.1.85.10

84 bytes from 10.1.85.10 icmp_seq=1 ttl=60 time=52.381 ms
84 bytes from 10.1.85.10 icmp_seq=2 ttl=60 time=61.591 ms
84 bytes from 10.1.85.10 icmp_seq=3 ttl=60 time=55.314 ms
84 bytes from 10.1.85.10 icmp_seq=4 ttl=60 time=47.121 ms
84 bytes from 10.1.85.10 icmp_seq=5 ttl=60 time=43.770 ms

PC3> █
```

Рисунок 15 – Пинги с ПК3

```
PC4> ping 192.168.14.10

84 bytes from 192.168.14.10 icmp_seq=1 ttl=59 time=60.366 ms
84 bytes from 192.168.14.10 icmp_seq=2 ttl=59 time=57.359 ms
84 bytes from 192.168.14.10 icmp_seq=3 ttl=59 time=73.034 ms
84 bytes from 192.168.14.10 icmp_seq=4 ttl=59 time=68.669 ms
84 bytes from 192.168.14.10 icmp_seq=5 ttl=59 time=61.160 ms

PC4> ping 192.168.15.10

84 bytes from 192.168.15.10 icmp_seq=1 ttl=60 time=46.344 ms
84 bytes from 192.168.15.10 icmp_seq=2 ttl=60 time=47.551 ms
84 bytes from 192.168.15.10 icmp_seq=3 ttl=60 time=51.572 ms
84 bytes from 192.168.15.10 icmp_seq=4 ttl=60 time=42.447 ms
84 bytes from 192.168.15.10 icmp_seq=5 ttl=60 time=53.401 ms

PC4> ping 10.2.63.10

84 bytes from 10.2.63.10 icmp_seq=1 ttl=62 time=36.429 ms
84 bytes from 10.2.63.10 icmp_seq=2 ttl=62 time=32.622 ms
84 bytes from 10.2.63.10 icmp_seq=3 ttl=62 time=30.333 ms
84 bytes from 10.2.63.10 icmp_seq=4 ttl=62 time=37.468 ms
84 bytes from 10.2.63.10 icmp_seq=5 ttl=62 time=22.279 ms

PC4> ping 10.2.74.10

10.2.74.10 icmp_seq=1 ttl=64 time=0.001 ms
10.2.74.10 icmp_seq=2 ttl=64 time=0.001 ms
10.2.74.10 icmp_seq=3 ttl=64 time=0.001 ms
10.2.74.10 icmp_seq=4 ttl=64 time=0.001 ms
10.2.74.10 icmp_seq=5 ttl=64 time=0.001 ms

PC4> ping 10.1.85.10

84 bytes from 10.1.85.10 icmp_seq=1 ttl=61 time=43.154 ms
84 bytes from 10.1.85.10 icmp_seq=2 ttl=61 time=33.072 ms
84 bytes from 10.1.85.10 icmp_seq=3 ttl=61 time=32.040 ms
84 bytes from 10.1.85.10 icmp_seq=4 ttl=61 time=32.285 ms
84 bytes from 10.1.85.10 icmp_seq=5 ttl=61 time=33.150 ms

PC4> █
```

Рисунок 16 – Пинги с ПК4

```
PC5> ping 192.168.14.10

84 bytes from 192.168.14.10 icmp_seq=1 ttl=59 time=75.862 ms
84 bytes from 192.168.14.10 icmp_seq=2 ttl=59 time=63.908 ms
84 bytes from 192.168.14.10 icmp_seq=3 ttl=59 time=51.994 ms
84 bytes from 192.168.14.10 icmp_seq=4 ttl=59 time=66.127 ms
84 bytes from 192.168.14.10 icmp_seq=5 ttl=59 time=66.355 ms

PC5> ping 192.168.15.10

84 bytes from 192.168.15.10 icmp_seq=1 ttl=60 time=55.398 ms
84 bytes from 192.168.15.10 icmp_seq=2 ttl=60 time=52.665 ms
84 bytes from 192.168.15.10 icmp_seq=3 ttl=60 time=52.970 ms
84 bytes from 192.168.15.10 icmp_seq=4 ttl=60 time=52.249 ms
84 bytes from 192.168.15.10 icmp_seq=5 ttl=60 time=45.129 ms

PC5> ping 10.2.63.10

84 bytes from 10.2.63.10 icmp_seq=1 ttl=60 time=69.553 ms
84 bytes from 10.2.63.10 icmp_seq=2 ttl=60 time=51.710 ms
84 bytes from 10.2.63.10 icmp_seq=3 ttl=60 time=52.510 ms
84 bytes from 10.2.63.10 icmp_seq=4 ttl=60 time=46.293 ms
84 bytes from 10.2.63.10 icmp_seq=5 ttl=60 time=45.686 ms

PC5> ping 10.2.74.10

84 bytes from 10.2.74.10 icmp_seq=1 ttl=61 time=50.134 ms
84 bytes from 10.2.74.10 icmp_seq=2 ttl=61 time=33.258 ms
84 bytes from 10.2.74.10 icmp_seq=3 ttl=61 time=47.651 ms
84 bytes from 10.2.74.10 icmp_seq=4 ttl=61 time=33.425 ms
84 bytes from 10.2.74.10 icmp_seq=5 ttl=61 time=36.124 ms

PC5> ping 10.1.85.10

10.1.85.10 icmp_seq=1 ttl=64 time=0.001 ms
10.1.85.10 icmp_seq=2 ttl=64 time=0.001 ms
10.1.85.10 icmp_seq=3 ttl=64 time=0.001 ms
10.1.85.10 icmp_seq=4 ttl=64 time=0.001 ms
10.1.85.10 icmp_seq=5 ttl=64 time=0.001 ms

PC5> 
```

Рисунок 16 – Пинги с ПК5

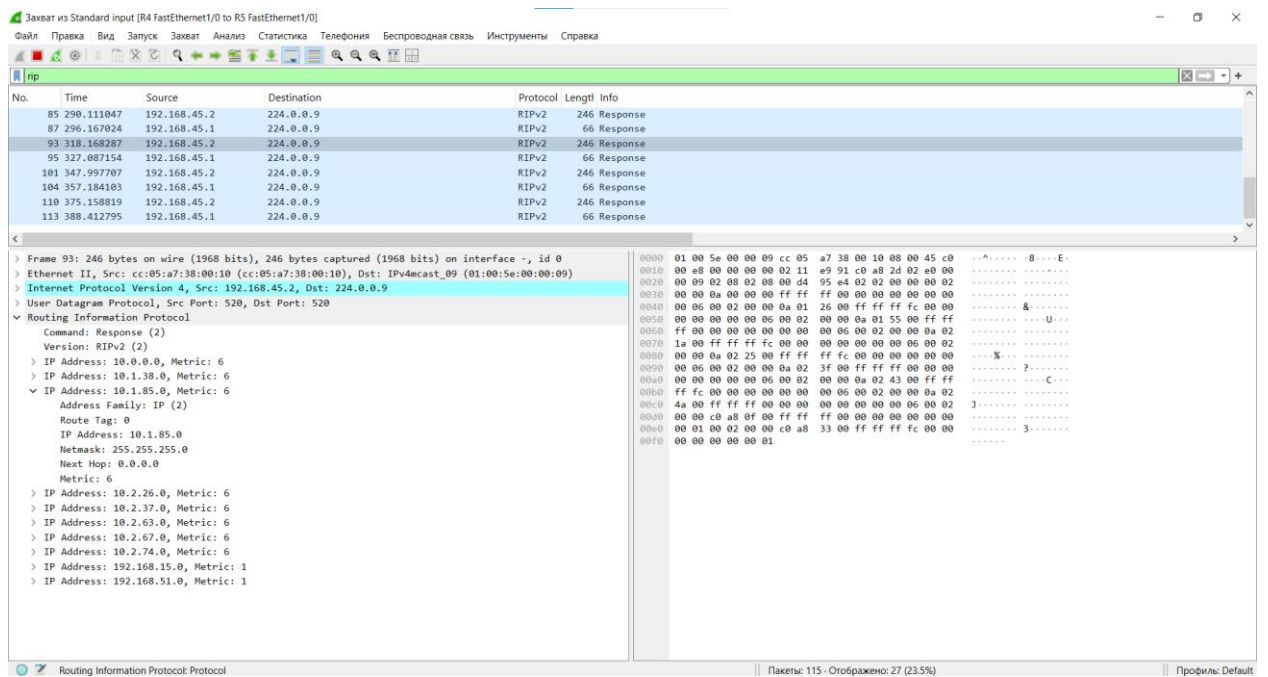


Рисунок 17 – Между R4 и R5

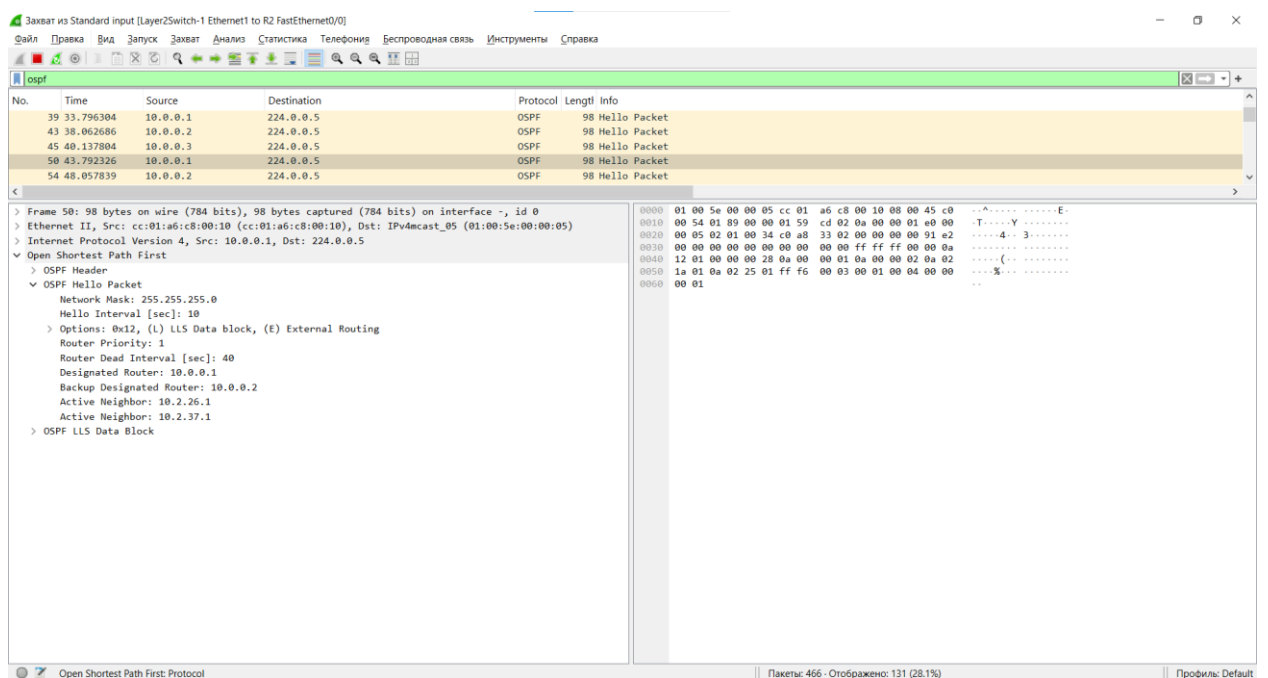


Рисунок 18 – Между R2 и Layer2Switch-1

Анализ пакета RIPv2

Метрики и параметры из пакета:

- **Протокол:** RIPv2 (работает поверх UDP, порт источника/назначения: 520).
- **Тип сообщения (Command):** Response (2) — пакет обновления (объявления) маршрутов.

- **Адрес отправителя (Source):** 192.168.45.2 (интерфейс роутера R5).
- **Адрес назначения (Destination):** 224.0.0.9 — групповой мультикаст-адрес протокола RIPv2.
- **Пример анонсируемой «внешней» сети (из OSPF):**
 - IP Address: 10.1.85.0
 - Netmask: 255.255.255.0
 - Metric: 6
- **Пример анонсируемой «родной» сети (RIP):**
 - IP Address: 192.168.15.0
 - Metric: 1

Описание происходящего:

На скриншоте зафиксирован процесс обмена маршрутной информацией RIPv2 на линке между R4 и R5. Маршрутизатор R5 отправляет пакет обновления Response на групповой адрес 224.0.0.9. Внутри сообщения передается список известных ему подсетей с указанием их масок и метрик (количества хопов), что позволяет соседнему роутеру R4 актуализировать свою таблицу маршрутизации.

В блоке данных прослеживается разница между типами маршрутов. Внутренние сети сегмента RIP (например, 192.168.15.0) анонсируются с метрикой 1. В то же время импортированные из OSPF сети (например, 10.1.85.0) имеют метрику 6. Это подтверждает успешную настройку редистрибуции: к базовой метрике перераспределения 5, заданной на пограничном R1, прибавился один хоп при прохождении через R5 до R4.

Анализ пакета OSPF

Метрики и параметры из пакета:

- **Протокол:** OSPF (работает напрямую поверх IP, номер протокола 89).
- **Тип сообщения (OSPF Packet Type):** Hello Packet (1) — служебный пакет для установления и поддержания соседства.
- **Адрес отправителя (Source):** 10.0.0.1 (интерфейс роутера R1).

- **Адрес назначения (Destination):** 224.0.0.5 — групповой мультикаст-адрес всех OSPF-маршрутизаторов.
- **Network Mask:** 255.255.255.0 (маска подсети интерфейса).
- **Hello Interval:** 10 секунд — период отправки Hello-пакетов.
- **Router Dead Interval:** 40 секунд — время ожидания перед объявлением соседа недоступным.
- **Designated Router (DR):** 10.0.0.1 (выбранный главный маршрутизатор сегмента).
- **Backup Designated Router (BDR):** 10.0.0.2 (резервный главный маршрутизатор).
- **Active Neighbors:** 10.2.26.1 (сосед R2) и 10.2.37.1 (сосед R3).

Описание происходящего:

На втором скриншоте представлен анализ периодического пакета OSPF Hello в многоадресном сегменте Area 0 (сеть коммутатора Layer2Switch-1). Маршрутизатор R1 транслирует данный пакет на мультикаст-адрес 224.0.0.5. Hello-пакеты отправляются каждые 10 секунд, а таймер удержания составляет 40 секунд.

Данные пакета подтверждают, что в общем сегменте успешно завершился процесс выборов выделенных маршрутизаторов: роль DR занял R1 (10.0.0.1), а роль BDR — R2 (10.0.0.2). В поле Active Neighbor зафиксировано успешное обнаружение соседних устройств R2 (10.2.26.1) и R3 (10.2.37.1). Идентичность параметров сетевой маски и таймеров во всех Hello-пакетах подтверждает отсутствие конфигурационных ошибок на стыке, что позволило роутерам перейти в состояние смежности FULL.